

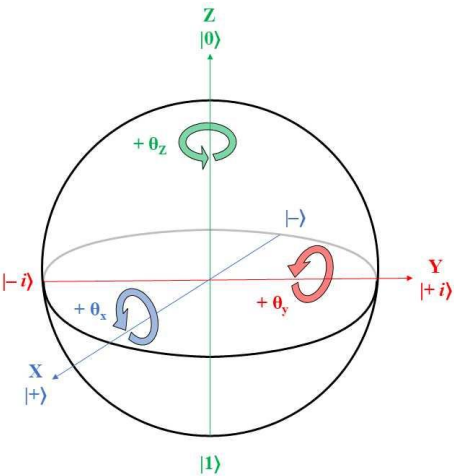
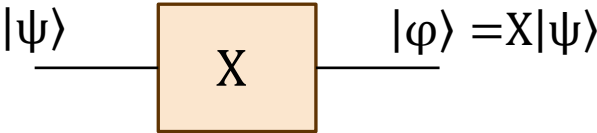
Introduction to Quantum Computing (L-3)

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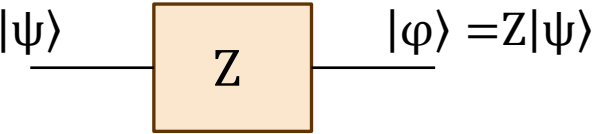
Single Qubit Quantum Gates (Recap from L2)

$X|0\rangle = |1\rangle \quad X|1\rangle = |0\rangle$

$X|\psi\rangle = X[\alpha|0\rangle + \beta|1\rangle] = \beta|0\rangle + \alpha|1\rangle$

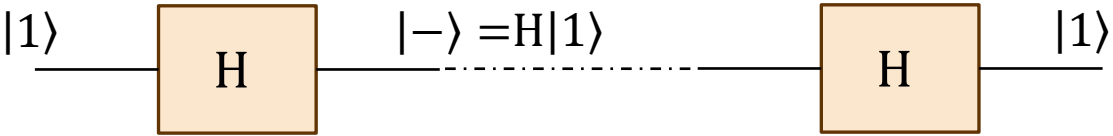
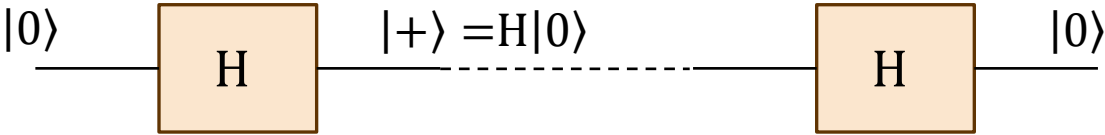


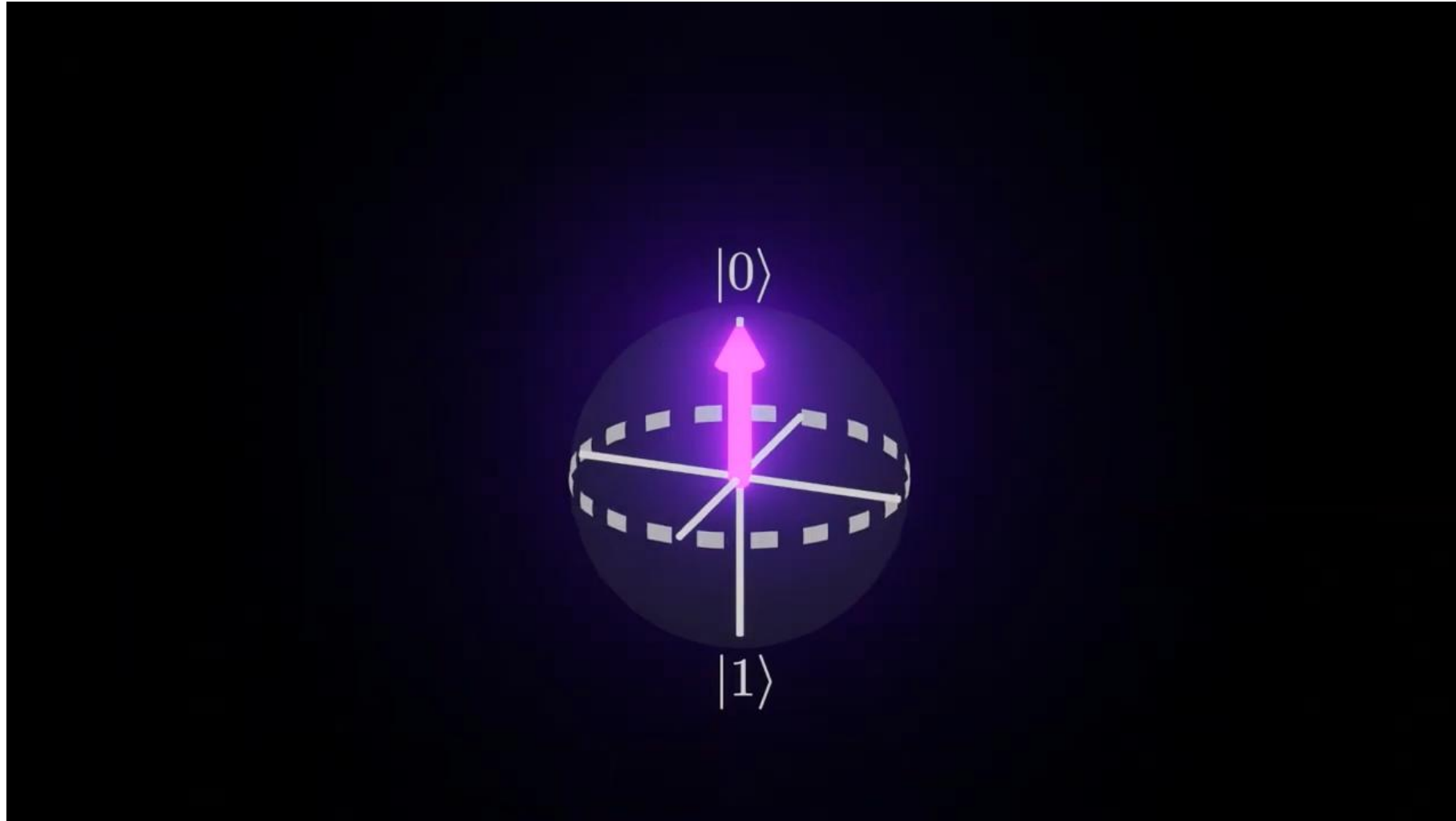
$Z|\psi\rangle = Z[\alpha|0\rangle + \beta|1\rangle]$
 $= \alpha Z|0\rangle + \beta Z|1\rangle = \alpha|0\rangle - \beta|1\rangle$



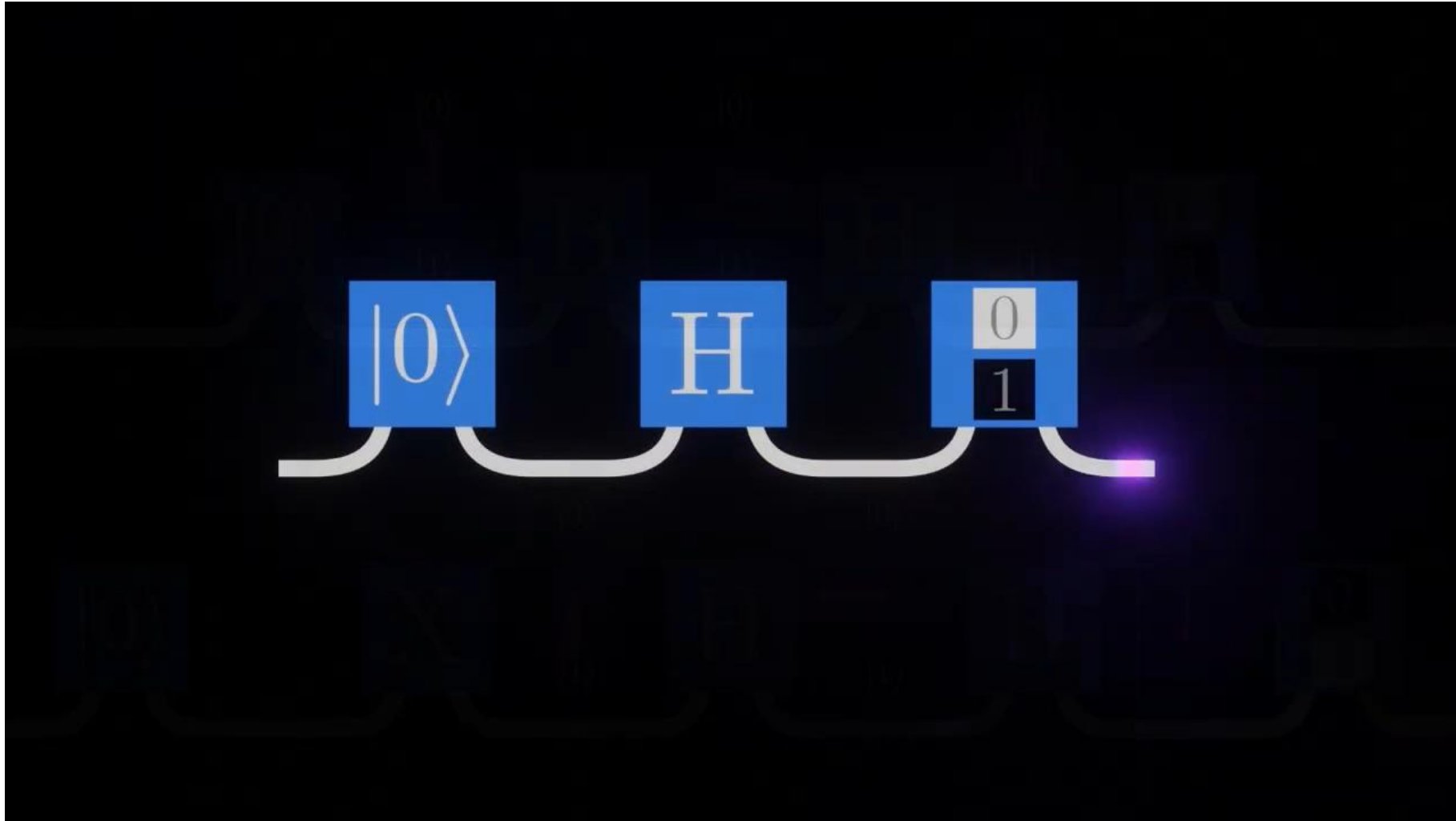
$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle$

$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$





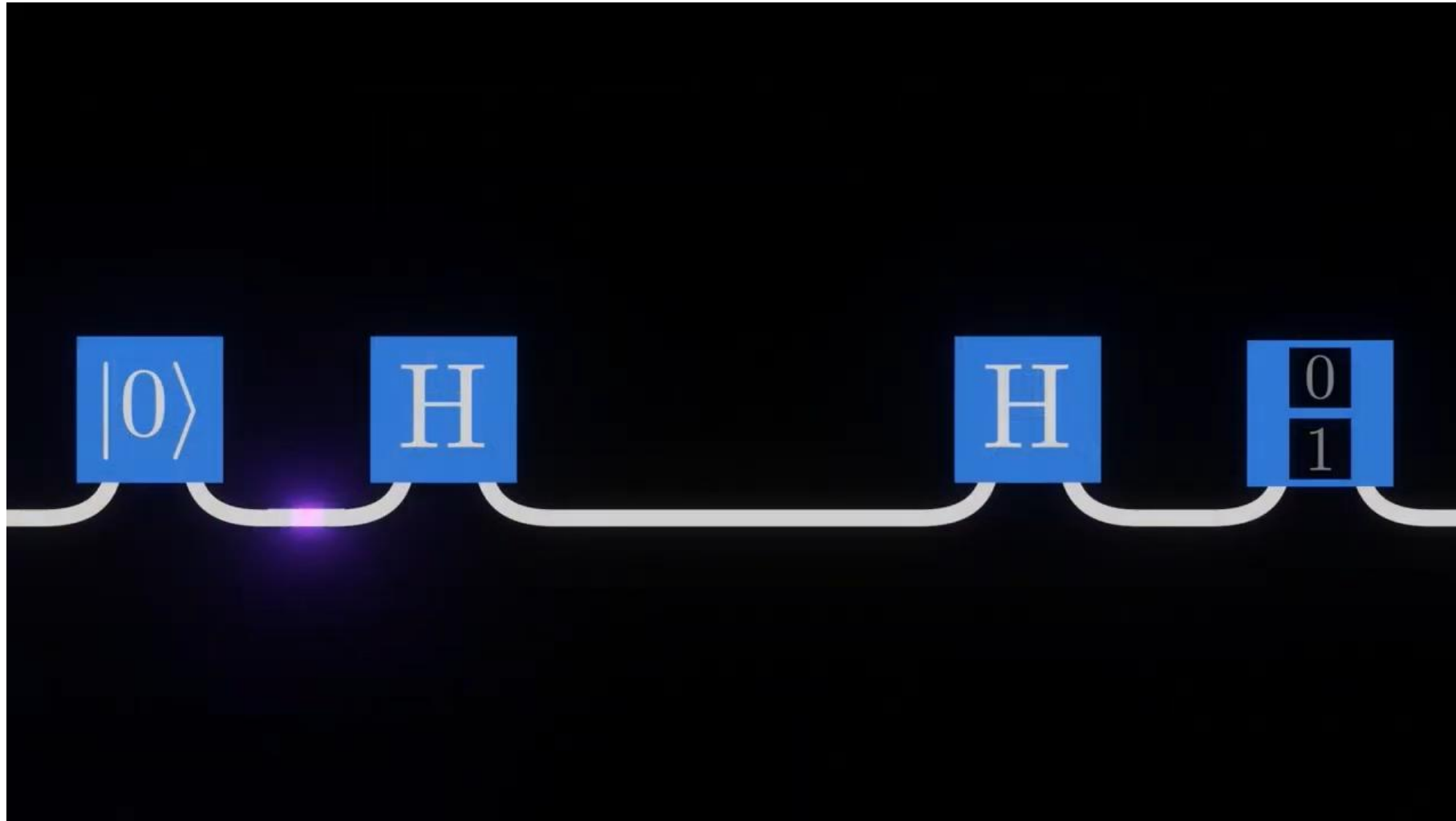
Bloch Sphere and Elementary Quantum Circuits



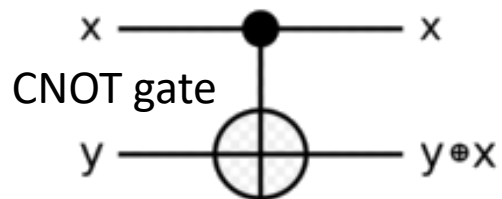
Hadamard Gate



Bloch Sphere and Elementary Quantum Circuits



The CNOT Gate



input		output	
x	y	x	y+x
0⟩	0⟩	0⟩	0⟩
0⟩	1⟩	0⟩	1⟩
1⟩	0⟩	1⟩	1⟩
1⟩	1⟩	1⟩	0⟩

input		output	
x	y	x	y+x
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

- Two qubits: control $\oplus$ target
- If control = $|1\rangle$, flip the target
- If control = $|0\rangle$, do nothing to target
- Unitary: $\text{CNOT}^\dagger = \text{CNOT}^{-1}$

$$|x\rangle = |0\rangle$$

$$|y\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|\psi\rangle_{\text{in}} = |0\rangle \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{|00\rangle + |01\rangle}{\sqrt{2}}$$

$\downarrow \text{CNOT}$

$$|\psi\rangle_{\text{out}} = \frac{|00\rangle + |01\rangle}{\sqrt{2}}$$

$$|x\rangle = |1\rangle$$

$$|y\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$|\psi\rangle_{\text{in}} = |1\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} = \frac{|10\rangle - |11\rangle}{\sqrt{2}}$$

$\downarrow \text{CNOT}$

$$|\psi\rangle_{\text{out}} = \frac{|11\rangle - |10\rangle}{\sqrt{2}}$$

$$|x\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|y\rangle = |0\rangle$$

$$|\psi\rangle_{\text{in}} = \frac{|0\rangle + |1\rangle}{\sqrt{2}} |0\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

$\downarrow \text{CNOT}$

$$|\psi\rangle_{\text{out}} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Introduction to Qiskit

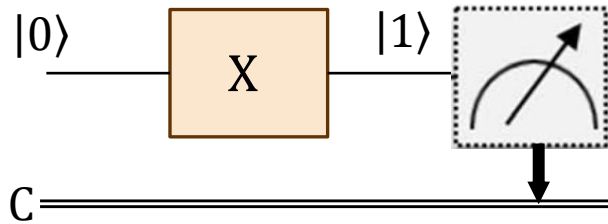
- **Qiskit is a Python-based quantum computing toolkit** that lets us write, visualize, and run quantum circuits using simple code
- **Works seamlessly inside Jupyter Notebook**, so circuits, math, plots, and explanations live together in one interactive workspace
- **Great for learning, not just theory**: we can see state vectors, Bloch spheres, circuit diagrams, and measurement results side by side
- **Current limitation in this class**: we will use simulators only — no pulse-level control, hardware calibration, or error-mitigation research
- **Why learn Qiskit?** It connects abstract quantum concepts directly to code and is widely used in academia, industry, and IBM Quantum resources
- **Official Qiskit YouTube channel** (tutorials, demos, short explainers):
<https://www.youtube.com/@qiskit>

Introduction to Qiskit

Subpackages	Function
<code>from qiskit import QuantumCircuit</code>	Creates and stores quantum circuits (qubits, gates, measurements).
<code>from qiskit.quantum_info import Statevector</code>	Represents and computes the exact quantum state (useful for Bloch spheres and theory).
<code>from qiskit_aer import Aer</code>	Provides high-performance simulators to run quantum circuits on a classical computer.
<code>from qiskit.visualization import plot_bloch_multivector, plot_histogram</code>	Draws quantum state visuals (Bloch sphere) and measurement outcome plots.
<code>import matplotlib.pyplot as plt</code>	General-purpose plotting library used to display figures generated by Qiskit.

Introduction to Qiskit

A simple X-gate on $|0\rangle$



How to visualize (not measure) state vectors at different points of a quantum circuit

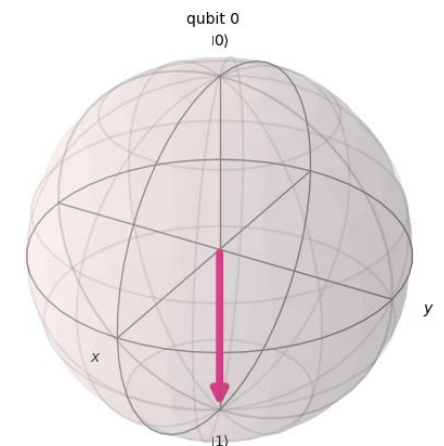
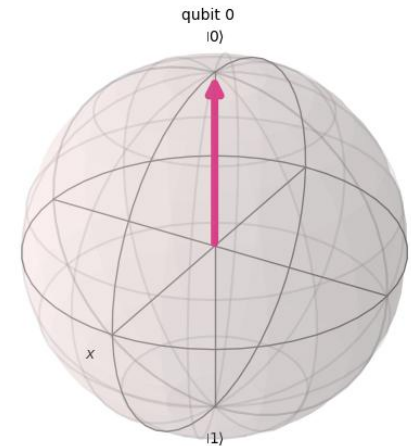
$$|\psi\rangle = a|0\rangle + b|1\rangle \quad \xrightarrow{\text{X}} \quad |\varphi\rangle = X|\psi\rangle$$

$$a = \sqrt{0.3}; b = \sqrt{0.7}$$

$$|\psi\rangle = \begin{bmatrix} \sqrt{0.3} \\ \sqrt{0.7} \end{bmatrix} \quad |\varphi\rangle = \begin{bmatrix} \sqrt{0.7} \\ \sqrt{0.8} \end{bmatrix}$$

How to represent qubits on a Bloch sphere

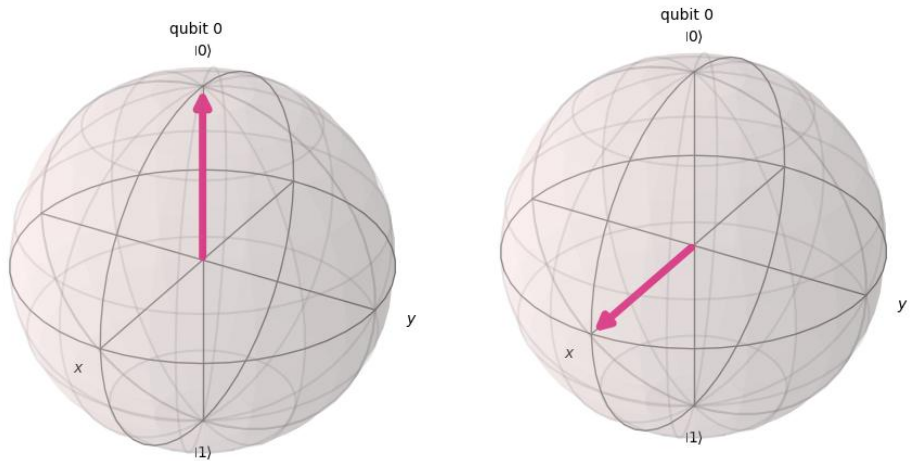
$$\begin{array}{l} |\psi\rangle = |0\rangle \\ |\psi\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{array} \quad \xrightarrow{\text{X}} \quad \begin{array}{l} |\varphi\rangle = X|\psi\rangle \\ |\varphi\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{array}$$



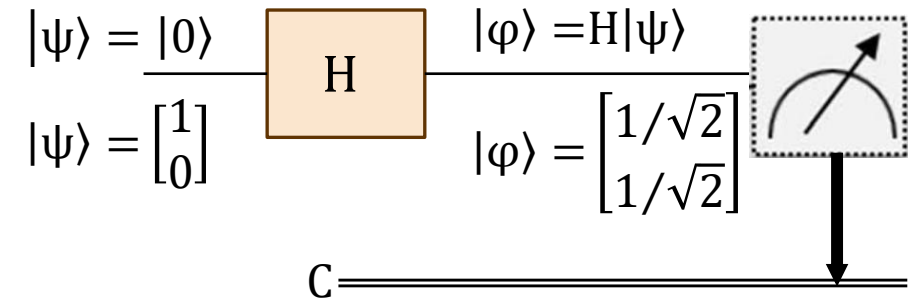
Introduction to Qiskit

Hadamard gate on $|0\rangle$

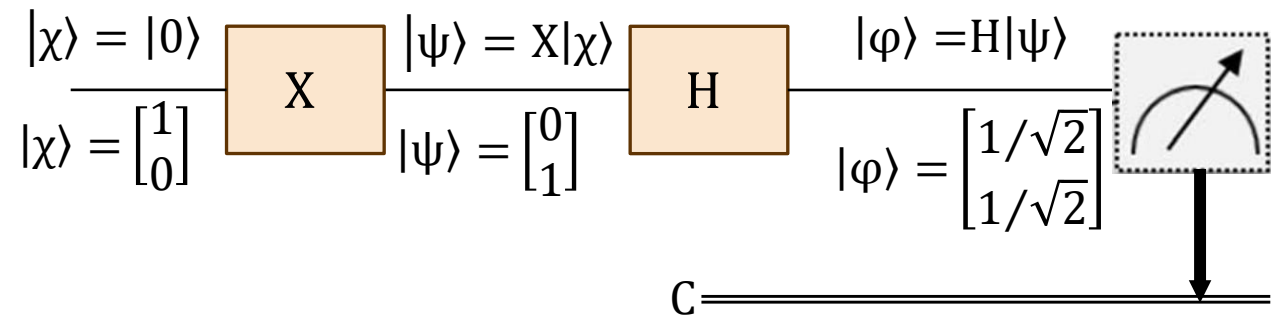
$$\begin{array}{lcl}
 |\psi\rangle = |0\rangle & \xrightarrow{\text{H}} & |\varphi\rangle = H|\psi\rangle \\
 |\psi\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} & & |\varphi\rangle = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}
 \end{array}$$



Measurement after a Hadamard gate on $|0\rangle$

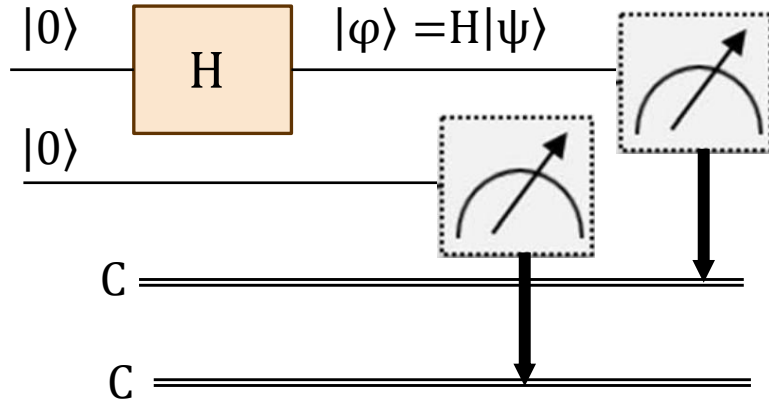


Measurement after a Hadamard gate on $|1\rangle$

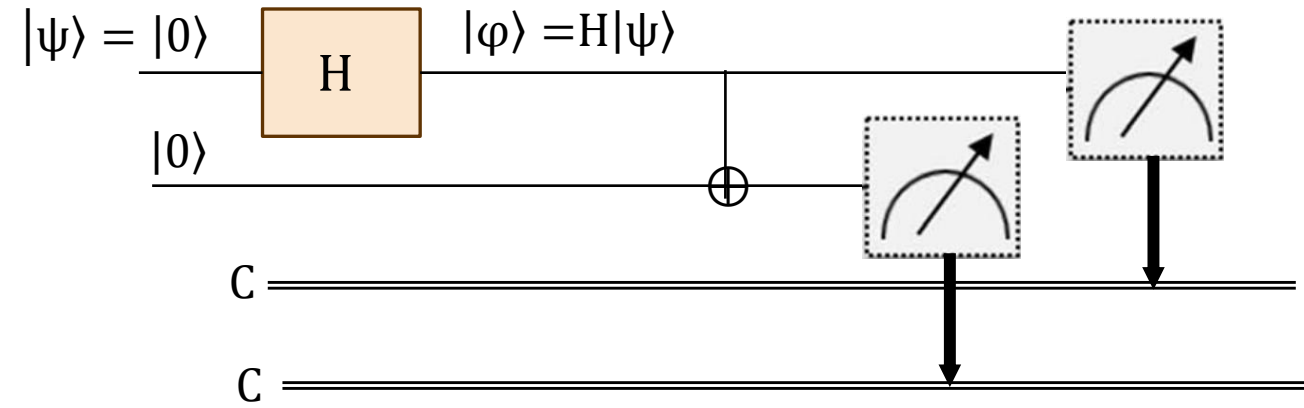
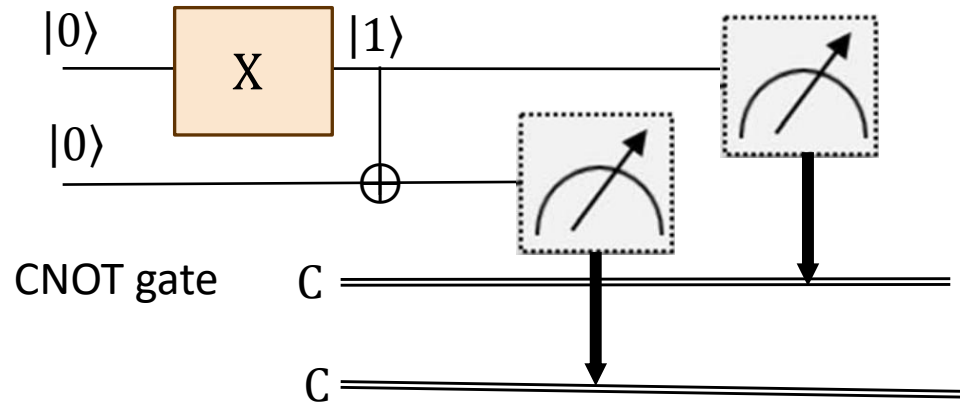


Introduction to Qiskit

Two qubits with no interaction between the qubits



Two qubits with interaction between the qubits -



Quiz